



[Project - Property Analysis]

BI Developer/Data Analyst Competition Sprint Part 1 - Property Data Engineering.

Introduction:

Welcome! The Property Analysis Competition Sprint is designed to follow an industry level best practice design of Business Intelligence project, it helps interns gain familiarity with Business Intelligence end to end solution, and also apply industry-level best practices in Business Intelligence and Data Analysis projects. While the instructions tell you what to do, they do not always necessarily tell you how to do it. This is deliberate as it is important for you to develop an independent drive to solve problems on your own. You should have a good idea where to start from your 6 weeks of training and the on-boarding tasks.

Download the Excel files from the link below.

[RawDataSet](#)

Part 1

Data Engineering: Design Data Warehouse, Data Model, ELT / ETL Data pipeline.

1. Click above RawDataSet link and download 3 datasets: AUS_SubCityDistrictState_Data, NSW_PropertyMedianValue, and NSW-Public-Schools-Master-Dataset.
2. Use Visual Studio and create SSIS package, design dataflows to extract data from AUS_SubCityDistrictState_Data, NSW_PropertyMedianValue and NSW-Public-Schools-Master-Dataset (csv), and load data into tables into your Data Warehouse, use "load_" as prefix to name all your data load tables in your Data Warehouse.
3. Follow Kimball dimensional modeling methodology, Design Snowflake Schema Model, or Star Schema Model including dimension and fact tables in Data Warehouse, use Bus Matrix and think about what dimension and fact tables you need to create. If you use the Star Schema Model you might need to consider using the factless fact table, do some research about the factless fact table.
4. Create another SSIS package and design dataflows to get Data extracted from load tables, then transformed and loaded into Snowflake Schema Model or Star Schema Model.
5. You need a "Category" as a dimension table (or degenerate dimension in fact table), in SSIS categorize Median Value into Category 0-750k, \$750k-\$1.5M, \$1.5M-\$2.5M,

\$2.5M+ and load the “Category” values into dimension table (or degenerate dimension in fact table) , this is a transformation of business requirements for Property Analysis.

Note: follow the same process as the onboarding task to submit your work to Mentor to review.